AI-ASSISTED CODING

ASSIGNMENT-4.3

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Task 1: Zero-Shot Prompting – Leap Year Check
Scenario
Zero-shot prompting involves giving instructions without providing examples.
Task Description
Use zero-shot prompting to instruct an AI tool to generate a Python function that:
• Accepts a year as input
• Checks whether the given year is a leap year
• Returns an appropriate result
Note: No input-output examples should be provided in the prompt.
Expected Output
• AI-generated leap year checking function
• Correct logical conditions
• Sample input and output
• Screenshot of AI-generated response (if required)

#PROMPT
generate code for Zero-shot prompting involves giving instructions without providing examples.Accepts a year as input.Checks whether the given year is a leap year .

#CODE
```
def is_leap_year(year):
    if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
        return True
    else:
        return False
# Get user input
year_input = int(input("Enter a year: "))
# Check if the year is a leap year and print the result
if is_leap_year(year_input):
    print(f"{year_input} is a leap year.")
else:
    print(f"{year_input} is not a leap year.")
```

#output

```
PROBLEMS    DEBUG CONSOLE    TERMINAL    PORTS    POSTMAN CONSOLE

 - Assisted coding/__pycache__/assinment 4.3.py/task1.py"
Enter a year: 2024
2024 is a leap year.
PS C:\Users\sidda\OneDrive\Documents\Desktop\AI - Assisted coding>
PS C:\Users\sidda\OneDrive\Documents\Desktop\AI - Assisted coding> & C:\Users\sidda\AppDat
a\Local\Programs\Python\Python314\python.exe "c:/Users/sidda/OneDrive/Documents/Desktop/AI
 - Assisted coding/__pycache__/assinment 4.3.py/task1.py"
Enter a year: 2025
2025 is not a leap year.
PS C:\Users\sidda\OneDrive\Documents\Desktop\AI - Assisted coding>
```

Task 2: One-Shot Prompting – Centimeters to Inches Conversion

Scenario

One-shot prompting guides AI using a single example.

Task Description

Use one-shot prompting by providing one input-output example to generate a Python function that:

- Converts centimeters to inches
- Uses the correct mathematical formula

Example provided in prompt:

Input: 10 cm → Output: 3.94 inches

Expected Output

- Python function with correct conversion logic
- Accurate calculation
- Sample test cases and outputs

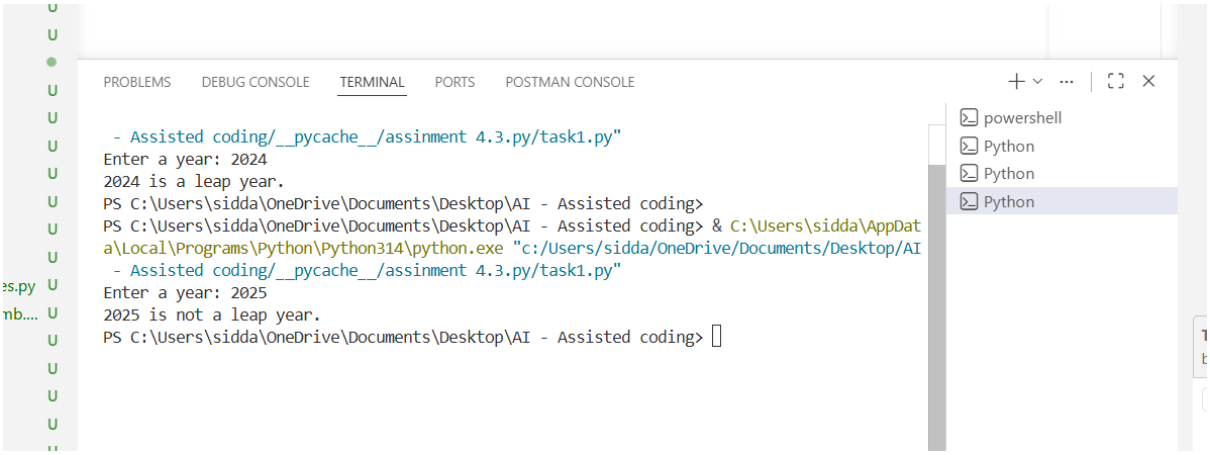
#prompt

generate code One-shot prompting guides AI using a single example.Converts centimeters to inches Uses the correct mathematical formula

#CODE

```
def cm_to_inches(cm):
    inches = cm / 2.54
    return inches
# Get user input
cm_input = float(input("Enter length in centimeters: "))
# Convert to inches and print the result
inches_output = cm_to_inches(cm_input)
print(f"{cm_input} cm is equal to {inches_output:.2f} inches.")
```

#OUTPUT



Task 3: Few-Shot Prompting – Name Formatting

Scenario

Few-shot prompting improves accuracy by providing multiple examples.

Task Description

Use few-shot prompting with 2–3 examples to generate a Python function that:

• Accepts a full name as input

• Formats it as “Last, First”

Example formats:

• "John Smith" → "Smith, John"

• "Anita Rao" → "Rao, Anita"

Expected Output

• Well-structured Python function

• Output strictly following example patterns

• Correct handling of names

• Sample inputs and outputs

#PROMPT

Few-shot prompting improves accuracy by providing multiple examples.Accepts a full name as input.Formats it as “Last, First".

#Code

```
def format_name(full_name):
    parts = full_name.split()
    if len(parts) >= 2:
        first_name = parts[0]
        last_name = parts[-1]
        return f"{last_name}, {first_name}"
    else:
        return "Please enter a full name with at least a first and last name."
# Get user input
name_input = input("Enter a full name: ")
# Format the name and print the result
```

formatted_name = format_name(name_input)

print("Formatted Name:", formatted_name)

#OUTPUT

```
basic text-proces.py  U
to find even numb....  U
                       U
df                     U
df                     U
df                     U
odf                    U

2025 is not a leap year.
PS C:\Users\sidda\OneDrive\Documents\Desktop\AI - Assisted coding> & C:\Users\sidda\AppDat
a\Local\Programs\Python\Python314\python.exe "c:/Users/sidda/OneDrive/Documents/Desktop/AI
 - Assisted coding/__pycache__/assinment 4.3.py/TASK3.PY"
Enter a full name: Velpula Siddartha
Formatted Name: Siddartha, Velpula
PS C:\Users\sidda\OneDrive\Documents\Desktop\AI - Assisted coding>
```

Task 4: Comparative Analysis – Zero-Shot vs Few-Shot

Scenario

Different prompt strategies may produce different code quality.

Task Description

• Use zero-shot prompting to generate a function that counts vowels in a string

• Use few-shot prompting for the same problem

• Compare both outputs based on:

o Accuracy

o Readability

o Logical clarity

Expected Output

• Two vowel-counting functions

• Comparison table or short reflection paragraph

• Conclusion on prompt effectiveness

#PROMPT

generate code Different prompt strategies may produce different code quality.Use zero-shot prompting to generate a function that counts vowels in a stringUse few-shot prompting for the same problem Compare both outputs based on: Accuracy ,Readability, Logical clarity

#code

```
def count_vowels_zero_shot(s):
    vowels = 'aeiouAEIOU'
    count = 0
    for char in s:
        if char in vowels:
            count += 1
    return count
```

```
# Few-shot prompting
def count_vowels_few_shot(s):
    vowels = 'aeiouAEIOU'
    count = 0
    for char in s:
        if char in vowels:
            count += 1
    return count
# Test the functions
test_string = "Hello World"
print("Zero-shot vowel count:", count_vowels_zero_shot(test_string))
print("Few-shot vowel count:", count_vowels_few_shot(test_string))
```

#OUTPUT

```
Enter a full name: Velpula Siddartha
Formatted Name: Siddartha, Velpula
PS C:\Users\sidda\OneDrive\Documents\Desktop\AI - Assisted coding> & C:\Users\sidda\AppDat
a\Local\Programs\Python\Python314\python.exe "c:/Users/sidda/OneDrive/Documents/Desktop/AI
 - Assisted coding/__pycache__/assinment 4.3.py/TASK4.PY"
Zero-shot vowel count: 3
Few-shot vowel count: 3
PS C:\Users\sidda\OneDrive\Documents\Desktop\AI - Assisted coding>
```

Task 5: Few-Shot Prompting – File Handling

Scenario

File processing requires clear logical understanding.

Task Description

Use few-shot prompting to generate a Python function that:

- Reads a .txt file
- Counts the number of lines in the file
- Returns the line count

Expected Output

- Working Python file-processing function
- Correct line count
- Sample .txt input and output
- AI-assisted logic explanation

#PROMPT

generate code File processing requires clear logical understanding.Use few-shot prompting to generate a Python function that:Reads a .txt file Counts the number of lines in the file Returns the line count

#CODE

```
def count_lines_in_file(file_path):
    try:
        with open(file_path, 'r') as file:
            lines = file.readlines()
            return len(lines)
    except FileNotFoundError:
        return "File not found. Please check the file path and try again."
# Get user input for file path
file_path_input = input("Enter the path to the .txt file: ")
# Count lines and print the result
line_count = count_lines_in_file(file_path_input)
print(f"The number of lines in the file is: {line_count}")
```

#OUTPUT